Regulation, efficiency, 3-phase transformers

Answers and worked solutions · Day 05 · Wed 28 Oct · pack M5

5. Practice

From your course files

Jun24 Q2c (past exam, 8 marks).

> This is worked in full as **Example 2** in Section 3 (with its circuit). Do it without looking first.
>
> i. Series branch $R = 0.08 + 0.06 = 0.14\ \Omega$ and $jX = j0.28\ \Omega$; shunt branch $250\ \Omega \parallel j14\ \Omega$ at the input.
>
> ii. Star: $V_1 = 16\,000/\sqrt{3} = 9237.6$ V. $Z = 250 \parallel j14 = 13.98\angle 86.79^\circ\ \Omega$, so $|I_1| = 9237.6/13.98 =$ **660.9 A** and $P = 3V_1^2/R_C =$ **1.024 MW**.
>
> iii. **×0.5, ×0.25, ×0.25, ×1.** All agree with the course answers.

2023 quiz Q5 (quiz, 5 marks).

> Hint: P_{IN} is given, so subtract the losses. Assume star, so $V_1 = 11\,700/\sqrt{3} = 6755$ V per phase.
>
> - Copper loss: $P_{CU} = 3I_2'^2R = 3(6.8)^2(8.5) =$ **1.179 kW**.
> - Iron loss: $P_{FE} = 3V_1^2/R_C = 3(6755)^2/100\,000 =$ **1.369 kW**.
> - Efficiency: $\eta = \dfrac{P_{IN} - P_{CU} - P_{FE}}{P_{IN}} = \dfrac{115 - 1.179 - 1.369}{115} =$ **97.79 %**.
>
> Course answer: 1.179 kW, 1.369 kW, 97.79 % (the answer sheet notes the factor of three). X and X_M are not needed.

2023 quiz Q6 (quiz, 9 marks).

> **(a)** $V_2' = 3300/\sqrt{3}\angle 0^\circ =$ **1.905∠0° kV**. $S = P/\text{pf} = 27\,000/0.78 = 34.62$ kVA, so
>
> $$|I_2'| = \frac{S}{3V_2'} = \frac{34\,615}{3(1905.3)} = 6.056\ \text{A}, \qquad I_2' = \mathbf{6.056\angle -38.74^\circ\ A}$$
>
> ($\cos^{-1} 0.78 = 38.74^\circ$, negative because lagging.)
>
> **(b)** Circuit: $18\,000\ \Omega \parallel j2300\ \Omega$ across V_1, then $4\ \Omega$ and $j35\ \Omega$ in series carrying I_2' to V_2' (the same layout as Section 1, with these values).
>
> $$V_1 = V_2' + I_2'(R + jX) = 2100 + 7.7\angle -25^\circ(4 + j35) = 2241.8 + j231.2 = 2.254\angle 5.889^\circ\ \text{kV}$$

Line voltage $= \sqrt{3} \times 2253.7 =$ **3.904 kV**.

> **NOTE ON THE OFFICIAL ANSWER.** The course gives $V_1 = 2.254\angle 5.889^\circ$ kV (which this matches exactly) but a line voltage of "3.883 kV". That does not follow from its own phase value: $\sqrt{3} \times 2253.7 = 3.904$ kV. I think 3.904 kV is correct and 3.883 is a slip. (The text version of the paper reads "21000 V"; the paper itself says 2100∠0° V.)

2023 quiz Q7 (quiz, 4 marks).

Model answer (following the course marking points):

- The number of primary turns sets the peak flux in the core for a given voltage and frequency, through the back-emf equation. The turns are chosen so that the operating peak flux density is below the saturation value of the core material.
- **Too few turns:** a higher peak flux density is needed to support the voltage. The core saturates, the magnetising current becomes very large (and distorted), and the winding may overheat.
- **Too many turns:** the core runs at a low flux density, so the core is under-used. The extra turns mean more winding length and resistance, so the copper loss is higher, the efficiency is lower, and the winding may overheat.

PQ 3.29 (practice questions).

Single phase, so there is no factor of 3. Use V_1 for the iron loss.

(a) SC: $R = 55/9.5^2 = 0.6094\ \Omega$, $|Z| = 27/9.5 = 2.842\ \Omega$,
$X = \sqrt{2.842^2 - 0.6094^2} = 2.776\ \Omega$. OC: $R_C = 240^2/45 = 1280\ \Omega$;
$S = 240 \times 1.2 = 288$ VA, $Q = \sqrt{288^2 - 45^2} = 284.5$ VAr,
$X_M = 240^2/284.5 = 202.5\ \Omega$.

(b) $V_2' =$ **240∠0° V**. $|I_2'| = 2400/240 = 10$ A and $\cos^{-1} 0.87 = 29.54^\circ$ lagging, so $I_2' =$ **10∠−29.54° A**.

(c) $V_1 = 240 + 10\angle -29.54^\circ (0.6094 + j2.776) = 259.0 + j21.1 =$ **259.9∠4.67° V**.

(d) $P_O = 2400 \times 0.87 =$ **2088 W**. $P_{CU} = 10^2 \times 0.6094 =$ **60.94 W**.
$P_{FE} = 259.85^2/1280 =$ **52.75 W**. $\eta = 2088/(2088 + 60.94 + 52.75) =$ **94.84 %**.

Course answers: 0.6094, 2.776, 1280, 202.5 Ω; 240∠0° V, 10∠−29.54° A; 259.9 V; 2088 W, 60.94 W, 52.77 W, 94.84 %. The iron loss differs only in the last digit because the official answer rounds V_1 first.

PQ 3.30 (practice questions).

Unity pf, and 3000 VA is the three-phase total. Assume star.

(a) $V_2' = 415/\sqrt{3} =$ **239.6∠0° V**. $I_2' = 3000/(3 \times 239.6) =$ **4.174∠0° A**.

(b) $V_1 = 239.6 + 4.174(0.95 + j8.5) = 243.57 + j35.48 = 246.1\angle 8.29^\circ$ V per phase. Line: $\sqrt{3} \times 246.1 =$ **426.3 V**.

(c) $P_{CU} = 3(4.174)^2(0.95) =$ **49.64 W**. $P_{FE} = 3(246.1)^2/3300 =$ **55.08 W**.

(d) $\eta = 3000/(3000 + 49.64 + 55.08) =$ **96.62 %**.

Course answers: 239.6∠0° V, 4.174∠0° A; 426.4 V [V_1 = 246.2 V]; 49.65 W, 55.10 W; 96.62 %. The last-digit differences come from rounding V_1.

PQ 3.31 (practice questions).

Convert the test values to per phase first. The OC test is at 415 V, not rated voltage.

(a) SC per phase: $V = 95/\sqrt{3} = 54.85$ V, $I = 500$ A, $P = 4000$ W. $R = 4000/500^2 =$ **0.01600 Ω**; $|Z| = 54.85/500 = 0.1097\ \Omega$, so $X =$ **0.1085 Ω**. OC per phase: $V = 415/\sqrt{3} = 239.6$ V, $P = 266.7$ W, $S = 239.6 \times 15 = 3594$ VA, $Q = 3584$ VAr. $R_C = 239.6^2/266.7 =$ **215.3 Ω**, $X_M = 239.6^2/3584 =$ **16.02 Ω**.

(b) $V_2' = 15\,000/\sqrt{3} =$ **8.660∠0° kV**. $S = 50/0.8 = 62.5$ MVA, $|I_2'| = 62.5\text{M}/(3 \times 8660) = 2406$ A, so $I_2' =$ **2.406∠−36.87° kA**. $V_1 = 8660 + 2406\angle -36.87^\circ(0.016 + j0.1085) = 8847.7 + j185.8 = 8.850\angle 1.20^\circ$ kV per phase, so the line value is **15.33 kV**.

(c) $P_{CU} = 3(2406)^2(0.016) = 0.2778$ MW. $P_{FE} = 3(8849.6)^2/215.3 = 1.091$ MW (the core runs at rated voltage here, far above the 415 V test). $\eta = 50/(50 + 0.2778 + 1.091) =$ **97.34 %**.

Course answers: 0.01600, 0.1085, 215.3, 16.02 Ω; 8.660∠0° kV, 2.406∠−36.87° kA, 15.33 kV [V_{PH} = 8.849 kV]; 97.34 %. All agree.

Extension: at rated load $P_{CU,FL} = 3(3849)^2(0.016) = 711$ kW. With the 1.091 MW iron loss, max η is at $x = 1.24$.

Jun24 Q4c(ii) (past exam, 3 marks).

- **Copper (winding) loss:** resistive heating of the primary and secondary windings by the load current. It is modelled by the series winding resistance (the primary resistance plus the referred secondary resistance). Because it depends on the square of the current, it rises with the square of the output.
- **Iron (core) loss:** hysteresis and eddy-current losses in the magnetic core, caused by the alternating flux. It is modelled by the core-loss resistance in the shunt branch

across the supply. It depends on the flux density and frequency, which are set by the supply voltage and frequency, so it is essentially constant whatever the output.
- Hence at light load the iron loss dominates, at heavy load the copper loss dominates, and the efficiency is greatest where the two are equal.

New questions

N1. Loss scaling and maximum efficiency [5 marks].

a. $P_{OUT} = 10\text{k} \times 0.8 = 8000$ W. $\eta = 8000/(8000 + 180 + 120) =$ **96.39 %**.

b. $x = 0.5$, so $P_{CU} = 0.5^2 \times 180 = 45$ W, while P_{FE} stays at 120 W. $P_{OUT} = 4000$ W. $\eta = 4000/(4000 + 45 + 120) =$ **96.04 %**.

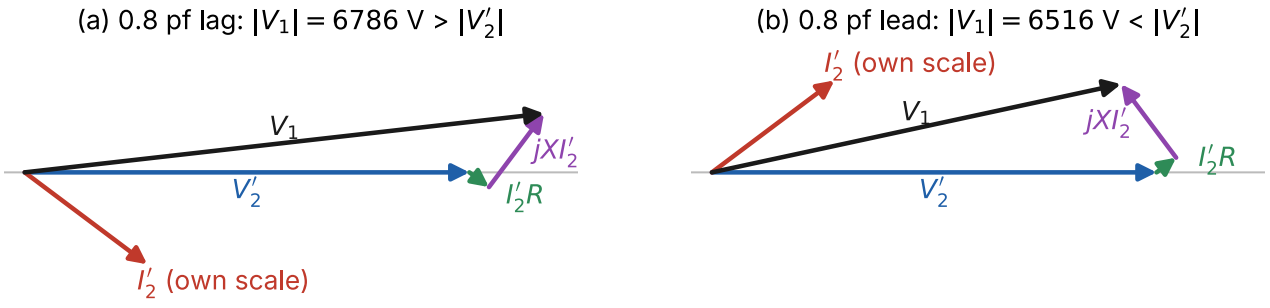
c. $x = \sqrt{120/180} = 0.8165$, so the load is **8.165 kVA**. At unity pf, $P_{OUT} = 8165$ W and the total loss is $2 \times 120 = 240$ W. $\eta_{max} = 8165/(8165 + 240) =$ **97.14 %**.

N2. Regulation with lagging and leading loads [6 marks].

$V_2' = 6600\angle 0°$ V and $|I_2'| = 50\,000/6600 = 7.576$ A.

a. $I_2' = 7.576\angle -36.87°$ A.
$V_1 = 6600 + 7.576\angle -36.87°(8 + j30) = 6784.8 + j145.5 =$ **6786 ∠1.23° V**.
Regulation $= (6786.4 - 6600)/6600 =$ **2.82 %**. Check:
$7.576(8 \times 0.8 + 30 \times 0.6) = 184.8$ V, which is 2.80 %.

b. $I_2' = 7.576\angle +36.87°$ A. $V_1 = 6600 + 7.576\angle 36.87°(8 + j30) =$ **6516 ∠1.92° V**.
Regulation $= (6515.8 - 6600)/6600 =$ **−1.28 %**. It is negative because the leading current rotates the jXI_2' drop so that it partly opposes V_2'. The output voltage is then higher than the no-load value (Prac 2 handout, Fig. 6).



New question 2, $V_2' = 6600\angle 0°$ V. Series drops drawn 6 times larger than scale.

Phasor diagrams for New question 2: (a) lagging load, V_1 larger than V_2′; (b) leading load, V_1 smaller than V_2′

c. Zero regulation needs $R\cos\phi = X\sin\phi$, so $\tan\phi = R/X = 0.2667$. That gives $\phi = 14.9^\circ$, a pf of **0.966 leading**. The exact phasor solution gives 0.961 leading.

N3. Delta-connected primary on open circuit [6 marks].

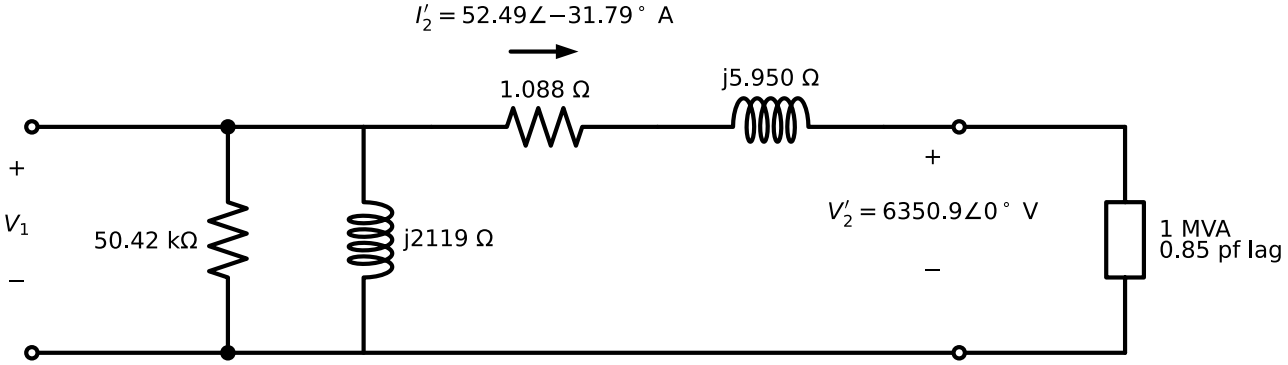
a. For delta, $V_1 = V_L = 11\,000$ V. $Z = 200\text{k} \parallel j30\text{k} = 29.67\angle 81.47^\circ$ kΩ, so $I_1 = 11\,000/Z =$ **0.371 ∠−81.5° A** per phase. Line current $I_L = \sqrt{3} \times 0.3708 =$ **0.642 A**. $P = 3V_1^2/R_C = 3(11\,000)^2/200\,000 =$ **1.815 kW**. $Q = 3V_1^2/X_M = 12.1$ kVAr. pf $= \cos 81.47^\circ =$ **0.148 lagging**.

b. Rated line current $= 500\text{k}/(\sqrt{3} \times 11\text{k}) = 26.24$ A. So $0.642/26.24 =$ **2.4 %**. This is typical of a large transformer (the Prac 2 handout quotes 2 to 4 %).

c. The model gives ×0.5, ×0.25, ×0.25, ×1. In practice the core runs further below saturation at half voltage, so X_M is larger and the current drops by **more** than half. The pf would also change. The model assumes constant R_C and X_M.

N4. Full chained analysis from test data [10 marks].

a. OC per phase: $V_1 = 11\,000/\sqrt{3} = 6350.9$ V and $P = 800$ W, so $R_C = 6350.9^2/800 =$ **50.42 kΩ**. $S = 6350.9 \times 3 = 19\,053$ VA and $Q = \sqrt{S^2 - P^2} = 19\,036$ VAr, so $X_M = V_1^2/Q =$ **2119 Ω**. SC per phase: $V_1 = 317.5$ V and $P = 3000$ W. $R = 3000/52.5^2 =$ **1.088 Ω**. $|Z| = 317.5/52.5 = 6.048\ \Omega$, so $X = \sqrt{6.048^2 - 1.088^2} =$ **5.950 Ω**.



Per-phase circuit for New question 4 with the part (a) parameters and the 1 MVA, 0.85 pf lagging load

b. $V_2' = 6350.9\angle 0^\circ$ V. Rated current $= 10^6/(\sqrt{3} \times 11\,000) = 52.49$ A, so $I_2' =$ **52.49 ∠−31.79° A**. $V_1 = 6350.9 + 52.49\angle -31.79^\circ(1.088 + j5.950) = 6568\angle 2.05^\circ$ V. Line voltage $= \sqrt{3} \times 6568.1 =$ **11.38 kV**. Regulation $= (6568.1 - 6350.9)/6350.9 =$ **3.42 %**.

c. $P_{OUT} = 0.85$ MW. $P_{CU} = 3(52.49)^2(1.088) = 8.995$ kW. $P_{FE} = 3(6568.1)^2/50\,417 = 2.567$ kW, using V_1. $\eta = 850/(850 + 8.995 + 2.567) =$ **98.66 %**.

d. The SC test was done at rated current, so $P_{CU,FL} = 9.0$ kW. $P_{FE} = 2.4$ kW at rated voltage. $x = \sqrt{2.4/9.0} = 0.516$, so the load is **≈ 517 kVA** (27.1 A). $\eta_{max} = 0.516 \times 850/(0.516 \times 850 + 2 \times 2.4) =$ **98.92 %**. Here the peak is well below rated load, which is the normal design for a power transformer.

N5. Short answer: no-load current and the Prac 2 evidence [4 marks].

a. The core has finite permeability, so a magnetising current is needed to supply the mmf that sets up the core flux. In the circuit this is the current through the magnetising reactance, and it lags the voltage by nearly 90°. The changing flux also produces hysteresis and eddy-current losses in the core. These need a real-power component of current, which is modelled by the core-loss resistance. Together these make up the no-load current, which is mostly reactive.

b. The peak core flux density is proportional to the applied voltage. Near rated voltage the core approaches saturation, and its permeability falls sharply. A much larger magnetising current is then needed for each extra increment of flux, so the magnetising reactance falls (from about 3980 Ω to about 366 Ω in Prac 2). The current rises far faster than the voltage and becomes distorted. The core-loss resistance changed much less, because iron loss depends on flux density and frequency rather than on permeability.

6. Quick recall check

Question

1. Copper loss ×0.25 (it goes as I^2). Iron loss ×1 (it stays constant at fixed V and f).
2. Iron loss uses V_1 ($P_{FE} = 3V_1^2/R_C$). Copper loss uses I_2' ($P_{CU} = 3I_2'^2 R$).
3. $P_{CU} = P_{FE}$, which occurs at $x = \sqrt{P_{FE}/P_{CU,FL}}$.
4. ×0.5, ×0.25, ×0.25, ×1.
5. $V_2' = 33\text{k}/\sqrt{3} = 19.05\angle 0°$ kV. Supply line voltage $= \sqrt{3}|V_1|$.